Teacher Wu Kuan | See It at a Glance #25 — The Law of Cosines

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1. The trap question

Sides 5 and 8 with a 60° angle between them. Find the third side.

2. The method

- 90° → Pythagoras $c^2 = a^2 + b^2$. Smaller angle → shorter c.
- Law of Cosines: $c^2 = a^2 + b^2 - 2ab \cos C$.
- $25 + 64 - 2 \cdot 5 \cdot 8 \cdot (1/2) = 49 \rightarrow c = 7$.
- Angle from 3 sides: $\cos C = (a^2 + b^2 - c^2) / 2ab$. Negative cosine → obtuse.

3. Parent check method

Ask: “What does the formula become when C = 90°?” ($\cos 90° = 0$ → Pythagoras.)

Practice

1. In triangle ABC, a = 3, b = 5, ∠C = 120°. Find c.
2. A triangle has sides 5, 7, 8. Find the angle opposite the side of length 7.
3. In triangle ABC, a = 4, b = 6, ∠C = 60°. Find c.
4. Is a triangle with sides 2, 3, 4 acute, right, or obtuse? Find the cosine of its largest angle.
5. A parallelogram has adjacent sides 4 and 6 with a 60° angle. Find both diagonals.

Answers

1. 7
2. 60°
3. $2\sqrt{7}$
4. Obtuse; cos = $-1/4$
5. $2\sqrt{7}$ and $2\sqrt{19}$

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